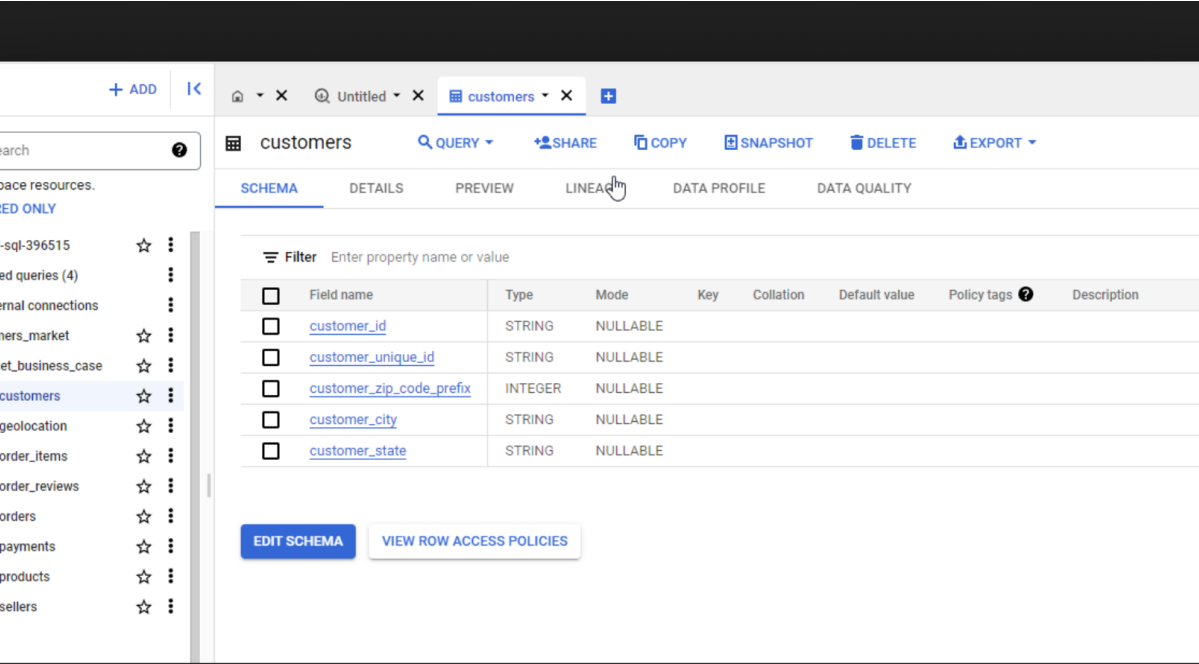


Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1.Data type of all columns in the "customers" table.

Ans :

The following screenshot is from BigQuery to check the column data types in the "customers" table:



Field name	Type	Mode	Key	Collation	Default value	Policy tags	Description
☐ customer_id	STRING	NULLABLE					
☐ customer_unique_id	STRING	NULLABLE					
☐ customer_zip_code_prefix	INTEGER	NULLABLE					
☐ customer_city	STRING	NULLABLE					
☐ customer_state	STRING	NULLABLE					

We ensure appropriate analysis and interpretation of the dataset by comprehending the data types of each table.

2. Get the time range between which the orders were placed.

Ans:

QUERY -

```
SELECT MIN(order_purchase_timestamp) AS start_time,  
MAX(order_purchase_timestamp) AS end_time  
FROM scaler-dsml-sql-396515.target_business_case.orders;
```

OUTPUT -

Query results			
JOB INFORMATION		RESULTS	JSON
		EXECUTION DETAILS	CHART
		PREVIEW	EXECUTION GRAPH
Row	start_time	end_time	
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	

The Query results: the start and end dates of the data i.e. from 4th September 2016 to 17th October 2018.

3.Count the Cities & States of customers who ordered during the given period.

Ans:

QUERY -

```
SELECT DISTINCT cust.customer_city, cust.customer_state,  
COUNT(ord.customer_id) AS order_count  
FROM target_business_case.customers cust JOIN  
target_business_case.orders ord USING(customer_id)  
GROUP BY cust.customer_city, cust.customer_state;
```

OUTPUT

Query results

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	CHART	PREVIEW
Row	customer_city	customer_state	order_count			
1	acu	RN	3			
2	ico	CE	8			
3	ipe	RS	2			
4	ipu	CE	4			
5	ita	SC	3			
6	itu	SP	136			
7	jau	SP	74			
8	luz	MG	2			
9	poa	SP	85			
10	uba	MG	53			
11	una	BA	5			
12	anta	RJ	4			
13	avai	SP	1			
14	bage	RS	65			
15	bodo	RN	1			
16	bora	SP	1			
17	buri	SP	10			
18	cacu	GO	8			

Here, we can observe that the city of Itu in the SP state alone has more orders than the combined total of the previous five cities. This is due to the fact that Brazil's most populous and prosperous state is Sao Paulo.

In-depth Exploration:

Is there a growing trend in the no. of orders placed over the past years?

Ans -

QUERY -

```
SELECT EXTRACT(YEAR FROM ord.order_purchase_timestamp) AS Year,
EXTRACT(MONTH FROM ord.order_purchase_timestamp) AS Month,
COUNT(DISTINCT ord.order_id) AS Order_count,
ROUND(SUM(p.payment_value),2) AS Revenue
FROM target_business_case.orders ord JOIN target_business_case.customers cust USING
(customer_id)
JOIN target_business_case.payments p USING(order_id)
GROUP BY Year, Month
ORDER BY Year, Month;
```

OUTPUT -

Query results

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS		CHART
Row	Year	Month	Order_count	Revenue		
1	2016	9	3	252.24		
2	2016	10	324	59090.48		
3	2016	12	1	19.62		
4	2017	1	800	138488.04		
5	2017	2	1780	291908.01		
6	2017	3	2682	449863.6		
7	2017	4	2404	417788.03		
8	2017	5	3700	592918.82		
9	2017	6	3245	511276.38		
10	2017	7	4026	592382.92		
11	2017	8	4331	674396.32		
12	2017	9	4285	727762.45		
13	2017	10	4631	779677.88		
14	2017	11	7544	1194882.8		
15	2017	12	5673	878401.48		
16	2018	1	7269	1115004.18		
Load more						

It is evident from the analysis of the amount of orders that e-commerce in Brazil is on the rise. With minor volatility, the number of purchases has shown an overall upward trend. It's crucial to remember that the order count alone does not reflect the rate of business expansion. We should also take revenue growth into account to have a more thorough perspective.

2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Ans:

QUERY -

```
SELECT EXTRACT(MONTH FROM order_purchase_timestamp) AS Month,
COUNT(DISTINCT order_id) AS Number_of_orders
FROM target_business_case.orders
GROUP BY Month
ORDER BY Month;
```

OUTPUT-

Query results			
JOB INFORMATION		RESULTS	JSON
Row	Month	Number_of_orders	
1	1	8069	
2	2	8508	
3	3	9893	
4	4	9343	
5	5	10573	
6	6	9412	
7	7	10318	
8	8	10843	
9	9	4305	
10	10	4959	
11	11	7544	
12	12	5674	

It is difficult to make firm predictions on seasonality trends given the small sample available. But according to the analysis, There are spikes and drops in the number of orders from March to August. Notably, there is a rise in orders in February and March, which falls around Brazil's Carnival season.

3. During what time of the day, do the Brazilian customers mostly place their orders?
(Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

Ans:

QUERY-

```
SELECT
CASE
  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN 'Dawn'
  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN 'Mornings'
  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN 'Afternoon'
  WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 19 AND 23 THEN 'Night'
END AS Day_Time,
COUNT(order_id) AS Number_of_orders
FROM target_business_case.orders
JOIN target_business_case.customers USING(customer_id)
GROUP BY Day_Time
ORDER BY Number_of_orders;
```

OUTPUT-

Query results		
JOB INFORMATION		RESULTS
Row	Day_Time	Number_of_orders
1	Dawn	5242
2	Mornings	27733
3	Night	28331
4	Afternoon	38135

The data revealed that Brazilian clients often place the majority of their orders during the day, particularly in the late afternoon and evening. This shows that users prefer to shop online during their free time or after finishing their everyday tasks.

Evolution of E-commerce orders in the Brazil region

1. Get the month on month no. of orders placed in each state.

Ans -

QUERY -

```
SELECT cust.customer_state,  
EXTRACT(MONTH FROM ord.order_purchase_timestamp) AS Month,  
COUNT(ord.order_purchase_timestamp) AS Number_of_orders  
FROM target_business_case.orders ord  
JOIN target_business_case.customers cust USING(customer_id)  
GROUP BY cust.customer_state, Month  
ORDER BY cust.customer_state, Month;
```

OUTPUT -

Query results				
JOB INFORMATION		RESULTS		JSON
Row	Month		Number_of_orders	
1	1		8069	
2	2		8508	
3	3		9893	
4	4		9343	
5	5		10573	
6	6		9412	
7	7		10318	
8	8		10843	
9	9		4305	
10	10		4959	
11	11		7544	
12	12		5674	

2.How are the customers distributed across all the states?

Ans -

QUERY -

```
SELECT customer_state,  
COUNT(DISTINCT customer_id) AS Number_of_customers  
FROM target_business_case.customers  
GROUP BY customer_state  
ORDER BY customer_state;
```

OUTPUT -

Query results			
JOB INFORMATION		RESULTS	JSON
Row	customer_state	Number_of_custome	EXECUTION DETA
1	SP	41746	
2	RJ	12852	
3	MG	11635	
4	RS	5466	
5	PR	5045	
6	SC	3637	
7	BA	3380	
8	DF	2140	
9	ES	2033	
10	GO	2020	
11	PE	1652	
12	CE	1336	
13	PA	975	
14	MT	907	
15	MA	747	
16	MS	715	
17	PB	536	
18	PI	495	

The data revealed that Brazilian clients often place the majority of their orders during the day, particularly in the late afternoon and evening. This shows that users prefer to shop online during their free time or after finishing their everyday tasks.

Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

Ans:

QUERY-

```
SELECT
  EXTRACT(MONTH FROM o.order_purchase_timestamp) AS Month,
  (
    (
      SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018 AND
        EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8 THEN
        p.payment_value END)
      -
      SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017 AND
        EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8 THEN
        p.payment_value END)
    )
    /
    SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017 AND
      EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8 THEN
      p.payment_value END)
  ) * 100 AS percent_increase
FROM target_business_case.orders o JOIN target_business_case.payments p USING(order_id)
WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018) AND
  EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
GROUP BY month
ORDER BY month;
```

OUTPUT -

Query results				
JOB INFORMATION		RESULTS		JSON
Row	Month		percent_increase	
1		1	705.1266954171...	
2		2	239.9918145445...	
3		3	157.7786066709...	
4		4	177.8407701149...	
5		5	94.62734375677...	
6		6	100.2596912456...	
7		7	80.04245463390...	
8		8	51.60600520477...	

Including only the months of January through August, The largest percentage rise can be seen when looking at the increase month-by-month; it is followed by February and April.

2. Calculate the Total & Average value of order price for each state.

Ans-

QUERY-

```
SELECT
  c.customer_state,
  ROUND(AVG(i.price), 2) AS Avg_price,
  ROUND(SUM(i.price), 2) AS total_price,
FROM target_business_case.customers c
JOIN target_business_case.orders o USING(customer_id)
JOIN target_business_case.order_items i USING(order_id)
GROUP BY c.customer_state
ORDER BY c.customer_state;
```

OUTPUT -

Query results			
JOB INFORMATION		RESULTS	JSON
EXECUTION DETAILS			
Row	customer_state	Avg_price	total_price
1	SP	109.65	5202955.05
2	RJ	125.12	1824092.67
3	MG	120.75	1585308.03
4	RS	120.34	750304.02
5	PR	119.0	683083.76
6	SC	124.65	520553.34
7	BA	134.6	511349.99
8	DF	125.77	302603.94
9	GO	126.27	294591.95
10	ES	121.91	275037.31
11	PE	145.51	262788.03
12	CE	153.76	227254.71
13	PA	165.69	178947.81
14	MT	148.3	156453.53
15	MA	145.2	119648.22
16	MS	142.63	116812.64
17	PB	191.48	115268.08
18	PI	160.36	86914.08

3. Calculate the Total & Average value of order freight for each state.

Ans:

QUERY -

```
SELECT
  c.customer_state,
  ROUND(AVG(i.freight_value), 2) AS Avg_price,
  ROUND(SUM(i.freight_value), 2) AS total_price,
FROM target_business_case.customers c
JOIN target_business_case.orders o USING(customer_id)
JOIN target_business_case.order_items i USING(order_id)
GROUP BY c.customer_state
ORDER BY c.customer_state;
```

OUTPUT -

Query results

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS
Row	customer_state	Avg_price	total_price	
1	AC	40.07	3686.75	
2	AL	35.84	15914.59	
3	AM	33.21	5478.89	
4	AP	34.01	2788.5	
5	BA	26.36	100156.68	
6	CE	32.71	48351.59	
7	DF	21.04	50625.5	
8	ES	22.06	49764.6	
9	GO	22.77	53114.98	
10	MA	38.26	31523.77	
11	MG	20.63	270853.46	
12	MS	23.37	19144.03	
13	MT	28.17	29715.43	
14	PA	35.83	38699.3	
15	PB	42.72	25719.73	
16	PE	32.92	59449.66	
17	PI	39.15	21218.2	
18	PR	22.52	117051.68	

Despite having the highest total price value and total freight value among all states, So Paulo (SP) has the lowest average price value and average freight value. The state of Roraima (RR), on the other side, has the greatest average freight value.

Analysis based on sales, freight and delivery time.

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order. Do this in a single query.

Ans:

QUERY -

```
SELECT
    order_id,
    DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY)
    AS delivery_time,
    DATE_DIFF(order_estimated_delivery_date, order_purchase_timestamp, DAY)
    AS estimated_delivery_days,
    DATE_DIFF(order_estimated_delivery_date, order_delivered_customer_date, DAY)
    AS diff_estimated_delivery
FROM target_business_case.orders
WHERE DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) IS NOT
NULL
ORDER BY delivery_time;
```

OUTPUT -

Query results

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	CHART	PREVIEW
Row	order_id	delivery_time	estimated_delivery_c	diff_estimated_delive		
1	e65f1eeee1f52024ad1dcd034...	0	10	9		
2	bb5a519e352b45b714192a02f...	0	26	25		
3	434cecee7d1a65fc65358a632...	0	20	19		
4	d3ca7b82c922817b06e5ca211...	0	12	11		
5	1d893dd7ca5f77ebf5f59f0d20...	0	10	10		
6	d5fbedc85190ba88580d6f82...	0	8	7		
7	79e324907160caea526fd8b94...	0	9	8		
8	38c1e3d4ed6a13cd0cf612d4c...	0	17	16		
9	8339b608be0d84fca9d8da68b...	0	28	27		
10	f349cdb62f69c3fae5c4d7d3f3...	0	13	12		
11	f3c6775ba3d2d9fe2826f93b71...	0	12	11		
12	b70a8d75313560b4acf607739...	0	10	9		
13	21a8ffca665bc7a1087d31751...	0	12	11		
14	44558a1547e448b41c48c4087...	1	7	5		
15	3bfd703ce884b8a0a65e63f2d...	1	7	5		

2. Find out the top 5 states with the highest & lowest average freight value.

Ans:

QUERY-

Top 5 Highest Average freight value

```
SELECT c.customer_state,
       ROUND(AVG(i.freight_value), 2) AS avg_freight_value
FROM target_business_case.customers c
JOIN target_business_case.orders o USING(customer_id)
JOIN target_business_case.order_items i USING(order_id)
GROUP BY c.customer_state
ORDER BY avg_freight_value desc
LIMIT 5;
```

OUTPUT -

Query results

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS
Row	customer_state	avg_freight_value		
1	RR	42.98		
2	PB	42.72		
3	RO	41.07		
4	AC	40.07		
5	PI	39.15		

QUERY -

Top 5 Lowest Average freight value

```
SELECT c.customer_state,
       ROUND(AVG(i.freight_value), 2) AS avg_freight_value
FROM target_business_case.customers c
JOIN target_business_case.orders o USING(customer_id)
JOIN target_business_case.order_items i USING(order_id)
GROUP BY c.customer_state
ORDER BY avg_freight_value
```

OUTPUT -

Query results

JOB INFORMATION		RESULTS	JSON	EXECU
Row	customer_state	avg_freight_value		
1	SP	15.15		
2	PR	20.53		
3	MG	20.63		
4	RJ	20.96		
5	DF	21.04		

3. Find out the top 5 states with the highest & lowest average delivery time.

Ans -

QUERY - Top 5 lowest average delivery time

```
SELECT c.customer_state,
       ROUND(AVG(DATE_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp,
DAY)), 2)
       AS Avgtime_to_delivery,
FROM target_business_case.customers c
JOIN target_business_case.orders o USING(customer_id)
JOIN target_business_case.order_items i USING(order_id)
GROUP BY c.customer_state
ORDER BY Avgtime_to_delivery
LIMIT 5;
```

OUTPUT -

Query results

JOB INFORMATION		RESULTS	JSON	EXECUTION
Row	customer_state	Avgtime_to_delivery		
1	SP	8.26		
2	PR	11.48		
3	MG	11.52		
4	DF	12.5		
5	SC	14.52		

Top 5 Highest Delivery time

```
SELECT c.customer_state,
       ROUND(AVG(DATE_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp,
                           DAY)), 2)
       AS Avgtime_to_delivery,
FROM target_business_case.customers c
JOIN target_business_case.orders o USING(customer_id)
JOIN target_business_case.order_items i USING(order_id)
GROUP BY c.customer_state
ORDER BY Avgtime_to_delivery desc
LIMIT 5;
```

OUTPUT

Query results

JOB INFORMATION		RESULTS	JSON	EXECUT
Row	customer_state	Avgtime_to_delivery		
1	RR	27.83		
2	AP	27.75		
3	AM	25.96		
4	AL	23.99		
5	PA	23.3		

The study finds a marginally positive link between mean freight value and delivery time. Roraima (RR) has the highest mean freight value, while So Paulo (SP) has the lowest.

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

Ans:

\

QUERY -

```
SELECT c.customer_state,
       ROUND(AVG(DATE_DIFF(o.order_estimated_delivery_date, o.order_delivered_customer_date,
DAY)), 2)
       AS fast_estimated_delivery
FROM target_business_case.customers c
JOIN target_business_case.orders o USING(customer_id)
JOIN target_business_case.order_items i USING(order_id)
GROUP BY c.customer_state
ORDER BY fast_estimated_delivery desc
LIMIT 5;
```

OUTPUT-

Query results			
JOB INFORMATION		RESULTS	JSON
Row	customer_state	fast_estimated_deliv	EXECUT
1	AC	20.01	
2	RO	19.08	
3	AM	18.98	
4	AP	17.44	
5	RR	17.43	

Analysis based on the payments:

1. Find the month on month no. of orders placed using different payment types.

Ans:

QUERY-

```
SELECT
  p.payment_type,
  EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month,
  COUNT(DISTINCT o.order_id) AS order_count
FROM target_business_case.orders o
JOIN target_business_case.payments p USING(order_id)
GROUP BY p.payment_type, month
ORDER BY p.payment_type, month;
```

OUTPUT -

Query results

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	
Row	payment_type	month	order_count		
1	UPI	1	1715		
2	UPI	2	1723		
3	UPI	3	1942		
4	UPI	4	1783		
5	UPI	5	2035		
6	UPI	6	1807		
7	UPI	7	2074		
8	UPI	8	2077		
9	UPI	9	903		
10	UPI	10	1056		
11	UPI	11	1509		
12	UPI	12	1160		
13	credit_card	1	6093		
14	credit_card	2	6582		
15	credit_card	3	7682		
16	credit_card	4	7276		
17	credit_card	5	8308		
18	credit_card	6	7248		

The most common payment method is credit card transactions, followed by UPI.

Transactions made with a debit card are the least recommended. Notably, compared to other payment methods, credit card transactions are growing quickly. This trend may be attributed to advantages like buy now, pay later or cashback rewards offered by credit cards.

2. Find the no. of orders placed on the basis of the payment installments that have been paid.

Ans:

QUERY -

```
SELECT
    p.payment_installments,
    COUNT(o.order_id) AS order_count
FROM target_business_case.orders o
JOIN target_business_case.payments p USING(order_id)
GROUP BY p.payment_installments
ORDER BY order_count desc;
```

OUTPUT -

Query results

JOB INFORMATION		RESULTS	JSON
Row	payment_installment	order_count	
1	1	52184	
2	2	12353	
3	3	10392	
4	4	7056	
5	10	5292	
6	5	5209	
7	8	4239	
8	6	3898	
9	7	1620	
10	9	638	
11	12	133	
12	15	74	
13	18	27	
14	11	22	
15	24	18	
16	20	17	
17	13	15	
18	14	15	

According to the data, the majority of orders only have one payment instalment (at the most). The highest instalments is 24, which are connected to 18 orders.

For businesses to optimise their payment operations and satisfy client preferences, they must have a thorough understanding of payment forms and instalment preferences.

ACTIONABLE INSIGHTS -

- According to the data, SP has a lot more orders than other states put together. This suggests that there is room for advancement in the other states. By concentrating on these states, you may boost sales and grow your clientele.
- Sales fluctuate seasonally, with holiday seasons seeing the biggest increases. To take advantage of these peak times and improve customer satisfaction, Target can organise their marketing and sales activities accordingly. This will lead to an increase in overall revenues.
- Enhancing delivery times in locations where deliveries take longer can boost customer satisfaction and promote repeat business. The key to accomplishing this is streamlining logistics and putting in place effective

shipping procedures.

- The data shows that orders decreased in September and October. By providing discounts or other incentives during off-peak times, businesses can encourage customers to make purchases and increase sales.
- States with high order counts include SP and RJ. It is advised to concentrate on customer retention methods in order to further increase revenue and promote brand loyalty.

RECOMMENDATIONS -

- To shorten delivery times and increase customer satisfaction, optimise logistics and shipping procedures. This entails streamlining shipping processes, fine-tuning shipping routes, and forming alliances with reputable courier firms.
- Utilise tactics for customer retention to promote loyalty and repeat business. Through loyalty programs, Referral Incentives
- Harness the power of social media platforms and influencers to endorse our products, boosting brand recognition, particularly in Brazil, where they wield considerable sway over purchase choices.
- Elevate the customer service encounter through the implementation of chat support services and the commitment to swift and proficient responses to customer questions.
- Analyse pricing and shipping costs to maintain a competitive edge in the market while optimising revenue and profitability. Evaluate the possibility of raising prices or modifying shipping fees as necessary.
- Foster partnerships with vendors to broaden our product range and enhance product quality, addressing a wide spectrum of customer desires and tastes.

CONCLUSION-

In summary, the examination of e-commerce data within the Brazilian market yields invaluable insights into customer purchasing behaviors, sales trajectories, preferred payment methods, and delivery satisfaction. Armed with this knowledge, enterprises can make well-informed choices and deploy tactics aimed at streamlining their processes and fostering expansion.